



**Yıldız Teknik Üniversitesi
Elektrik-Elektronik Fakültesi
Bilgisayar Mühendisliği Bölümü**

BLM 2642

**Bilgisayar Mühendisleri için
Diferansiyel Denklemler**

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Homework 2**

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Answer 1)

$$P(t) = 2 \\ Q(t) = e^t$$

$$\frac{dy}{dt} + 2y = e^t \quad y' + P(t)y = Q(t)$$

$$\mu = e^{\int P(t)} = e^{2t} \quad e^{2t} y = \frac{e^{3t}}{3} + C$$

$$\mu \cdot y = \int \mu Q(t)$$

$$e^{2t} \cdot y = \int e^{2t} \cdot e^t$$

$$\boxed{y = \frac{e^t}{3} + \frac{C}{e^{2t}}}$$

Answer 2)

$$\frac{dy}{dx} = xy^2 \quad \int y^{-2} \cdot dy = \int x \cdot dx$$

$$-y^{-1} = \frac{x^2}{2}$$

$$\boxed{y = -2/x^2}$$

Answer 3)

$$(x^2 + y)dx + (y^2 + x)dy = 0$$

$$Mdx + Ndy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \frac{\partial (x^2 + y)}{\partial y} = \frac{\partial (y^2 + x)}{\partial x}$$

$$\Rightarrow 1 = 1 \quad \text{ifade } \underline{\underline{\text{exact diff}}}$$

$$\int (x^2 + y) dx = \int (y^2 + x) dy$$

$$\frac{x^3}{3} + xy + c(x) = \frac{y^3}{3} + xy + c(y)$$

$$c(y) = \frac{y^3}{3} \quad \boxed{\frac{x^3}{3} + xy + \frac{y^3}{3} = K}$$

Answer b)

$$\frac{dy}{dx} + y = xy^2$$

$$V = y^{1-n}$$

$$V = y^{-1}$$

$$y' + p(x) \cdot y = Q(x) \cdot y^n$$

$$p(x) = 1 \quad Q(x) = x \quad n = 2$$

$$\frac{dv}{dx} = y^{-2} \frac{dy}{dx}$$

$$\frac{dv}{dx} \cdot y^2 = \frac{dv}{dx}$$

$$y = xy^2 + \frac{dv}{dx} \cdot y^2$$

$$y = y^2 \left(x + \frac{dv}{dx} \right)$$

$$\frac{1}{y} = x + \frac{dv}{dx}$$

$$V = x + \frac{dv}{dx}$$

$$p(x) = -1 \quad \mu = e^{\int -1} = e^{-x}$$

$$Q(x) = -x$$

$$\hookrightarrow \mu \cdot V = \int -1 \cdot \mu \Rightarrow e^x \cdot V = \int -x \cdot e^{-x} \Rightarrow e^{-x} (x+1) + C$$

$$e^x \cdot V = e^{-x} (x+1) + C \cdot e^x$$

$$V = x+1 + C \cdot e^x$$

$$\boxed{y = \frac{1}{x+1+Ce^x}}$$

Answer 5)

$$\frac{dy}{dt} = 2t$$

$$\int dy = \int 2t \cdot dt$$

$$y = t^2 + C$$

$$y(0) = 0 + C = 3$$

$$\underline{C = 3}$$

$$\boxed{y(t) = t^2 + 3}$$

Answer 6)

$$\frac{dT}{dt} = -k(T - T_a)$$

$$M = e^{\int p(x)} = e^{\int k} = e^{kt}$$

$$T' + kT = kT_a$$

$$y' + p(x)y = Q(x)$$

$$\hookrightarrow T' + p(k) \cdot T = Q(k)$$

$$p(k) = k$$

$$Q(k) = kT_a$$

$$M \cdot T = \int M \cdot Q(k)$$

$$e^{kt} \cdot T = \int e^{kt} \cdot k \cdot T_a$$

$$e^{kt} \cdot T = \cancel{k} \cdot T_a \cdot \frac{e^{kt}}{\cancel{k}} + C$$

$$e^{kt} \cdot T = T_a \cdot e^{kt} + C$$

$$T = T_a + \frac{C}{e^{kt}}$$

$$T(0) = T_a + C$$

$$C = T_0 - T_a$$

$$\boxed{T = T_a + \frac{T_0 - T_a}{e^{kt}}}$$

Answer 7)

$$\frac{dp}{dt} + p = e^{-2t}$$

$$y' + p(x) \cdot y = Q(x)$$

$$p' + A(x) \cdot p = B(t)$$

$$A(x) = 1 \quad B(t) = e^{-2t}$$

$$\mu = e^{\int A(x)} = e^{\int 1} = e^t$$

$$\mu A(x) = \int B(x) \cdot \mu$$

$$e^t \cdot p = \int e^{-2t} \cdot e^t$$

$$e^t \cdot p = -e^{-t} + C$$

$$p = -e^{-2t} + \frac{C}{e^t}$$

$$p_0 = -1 + C$$

$$C = p_0 + 1$$

$$p = e^{-2t} + \frac{(p_0 + 1)}{e^t}$$

Answer 8)

$$f(w) = w^2 + 2w + 3$$

$$w_{n+1} = w_n + a \cdot \frac{df}{dw}$$

$$w_{n+1} = w_n + 0.1(2w + 2)$$

$$w_{n+1} = 0.8w_n + 0.2$$

$$\begin{aligned} w_1 &= 2,2 \\ w_2 &= 1,56 \\ w_3 &= 1,048 \end{aligned}$$

$$\frac{df}{dw} = 2w + 2$$

$$a = 0.1$$

$$w_1 = 0.8 \cdot 3 - 0.2$$

$$w_1 = 2,2$$

$$w_2 = 0.8 \cdot 2,2 - 0.2$$

$$w_2 = 1,56$$

$$w_3 = 0.8 \cdot 1,56 - 0.2$$

$$w_3 = 1,048$$